

Finite Configurations Cannot Generate a Constant Trace in Rule 30

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For every right half with $R_0 = 0$ we determine the unique left half whose Rule-30 central trace is identically zero, and for every right half with $R_0 = 1$ the unique left half whose trace is identically one. The first is an alternating tail selected by the leading one of the right half; the second is a single checkerboard, the same for every right half. Every nonzero member of either fiber carries infinitely many ones, so the union of the two fibers contains exactly one finitely supported configuration, the zero row. For support radius w , the sharp maximum constant-prefix length is $2\lceil w/2 \rceil + 1$ when the initial center is zero and $2\lfloor w/2 \rfloor + 2$ when it is one. The maximum over both constants is therefore $w + 2$, attained by exactly 2^w configurations for even w and $2^w - 1$ for odd w . Hence no column of a nonzero finite Rule 30 orbit is eventually constant.

1 Introduction

Rule 30 is the binary cellular automaton on $\{0, 1\}^{\mathbb{Z}}$ whose local rule is the following.

$$F(x)_i = x_{i-1} \oplus (x_i \vee x_{i+1}) \tag{1}$$

It is left permutive: fixing x_i and x_{i+1} , the value $F(x)_i$ is a bijection of x_{i-1} .

Wolfram's Prize Problem 1 asks whether the center trace from the lone-seed configuration is eventually periodic [7]; Kopra records the broader finite-configuration version as open [4]. Jen proved that a finite Rule 30 orbit has at most one eventually periodic column [2, Thm. 2b][1, Prop. 3], while Kopra's width-two theorem does not decide whether that single exceptional column exists [3]. We settle the entire eventual-period-one subcase for every nonzero finite configuration, a class strictly broader than the lone seed. This is a direct partial result toward the center-column question, but it does not resolve Prize Problem 1 and leaves every nonconstant eventual period open.

The mechanism is an exact inversion: prescribing the right half and demanding an all-zero center forces the left half cell by cell, as a prefix-OR of the right half. A right half whose first one sits at depth 3, for instance, forces $L_1, L_2, \dots = 0, 0, 1, 0, 1, 0, 1, \dots$, with ones at every odd depth past the third. The uniqueness step is the left-permutive inverse argument of [5]. What makes the result Rule-30-specific is the OR in (1), which latches; the analogous claim is false for Rule 90 (Section 3). Figure 1 shows the forced left half and that contrast.

Two prior results bound what is new here. Jen gives necessary and sufficient conditions for a finite initial condition to generate an eventually-one temporal sequence [2, Thm. 7a]. Rule 30 meets that theorem's hypothesis $a_0 = 0, a_1 = a_4 = 1$ but none of its four alternatives: their coefficient conditions require, respectively, $a_6 = a_7 = 1, a_6 = 1, a_7 = 1$, or $a_5 = 1$, whereas Rule 30 has $a_5 = a_6 = a_7 = 0$. The necessity direction therefore already excludes an eventually-one column for every finite Rule 30 configuration. We do not claim novelty for that finite-support exclusion. The results here instead give both exact fixed-locus fibers, establish the Rule-30 zero exclusion explicitly, and supply sharp quantitative horizons and extremizer counts. Jen remarks that symmetric results hold for constant-zero sequences, but does not state a Rule-30 zero-fiber classification or any of these quantitative conclusions. Separately, inversion for nearest-neighbor rules has been solved for the *additive* class over $\mathbb{Z}_p$, by an operator analogous to backward integration [6], against a general predecessor problem that is NP-complete. Theorem 2 inverts a constraint imposed over all future time rather than one step, for a rule that is not additive.

2 The constant-trace fibers

The trace and the two halves of a row are written as follows.

$$c_t := \text{Tr}(x)_t = F^t(x)_0, \quad R_j := x_j, \quad L_j := x_{-j} \quad (j \geq 0) \quad (2)$$

Throughout, $\mathcal{F}$ denotes the finite-support configurations.

The trace sees only column 0, but under a left-permutive rule a discrepancy left of the origin propagates rightward along a diagonal and must eventually reach it. The classification rests on this one fact, which uses only locality and left permutivity, not the specific form of (1).

Lemma 1 (Triangular uniqueness). *If two configurations agree at every coordinate $i \geq 0$ and have the same central trace, then they are equal.*

Proof. If $-n$ is their rightmost differing coordinate, locality preserves agreement to its right through time n . Left permutivity preserves the discrepancy along $(-n, 0), (-n+1, 1), \dots, (0, n)$, contradicting equal traces. In its finite form, if the rows agree on $i \geq 0$ and their traces agree for $0 \leq t \leq r$, the same induction forces agreement at $-r, -r+1, \dots, -1$.

Uniqueness alone does not say what the forced left half looks like. The next result computes it: the rows with zero trace form one invariant class $\mathcal{C}_m$ for each position m of the first right-hand one, each class is carried to its predecessor by F , and the left half is a prefix-OR of the right. Figure 1(a) is a member of $\mathcal{C}_3$.

Theorem 2 (Zero-trace fiber). *For a prescribed right half with $R_0 = 0$, the zero-trace-compatible left half is unique. Suppose the positive half is nonzero, and let $m = \min\{j \geq 1 : R_j = 1\}$. The forced left half is then the following, with R_j for $j > m$ arbitrary.*

$$L_j = 0 \quad (j < m), \quad L_m = 1, \quad L_j = j \bmod 2 \quad (j > m) \quad (3)$$

If instead the positive half is zero, the unique row is zero. The same classification in prefix-OR form is the following.

$$L_{2k+1} = \bigvee_{j=1}^{2k+1} R_j \quad (k \geq 0) \quad (4)$$

$$L_{2k} = R_{2k} \wedge \neg \bigvee_{j=1}^{2k-1} R_j \quad (k \geq 1) \quad (5)$$

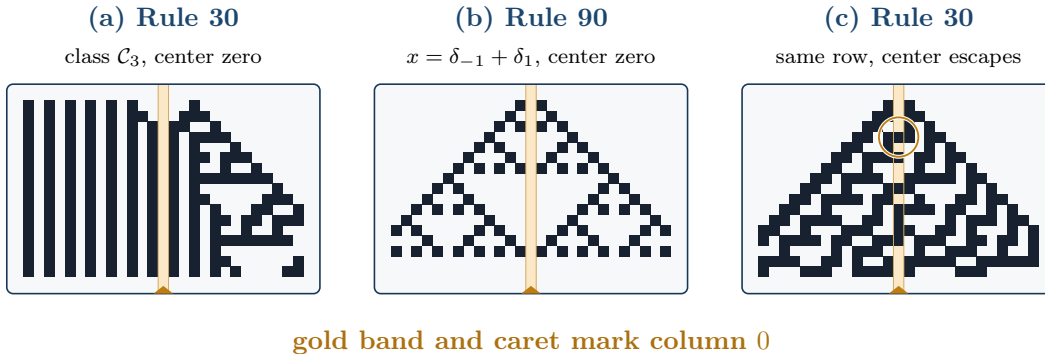


Figure 1: Time runs downward. In (a) the forced left tail alternates, hence is infinite, which is what Corollary 4 exploits. Panels (b) and (c) run one finite row under both rules; the difference is the OR latch in (1).

Proof. For $m \geq 1$, let $\mathcal{C}_m$ consist of rows satisfying these conditions. One Rule 30 step gives the three identities below.

$$\begin{aligned} x'_0 &= L_1 \oplus R_1 \\ R'_j &= R_{j-1} \oplus (R_j \vee R_{j+1}) \\ L'_j &= L_{j+1} \oplus (L_j \vee L_{j-1}) \end{aligned} \tag{6}$$

Suppose $m > 1$. These identities then give the following.

$$x'_0 = 0, \quad R'_j = L'_j = 0 \ (j < m - 1), \quad R'_{m-1} = L'_{m-1} = 1, \quad L'_m = m \bmod 2 \tag{7}$$

For $j > m$, including $j = m + 1$ because $L_m = 1$, the OR in the last update is one, so $L'_j = (1 - (j \bmod 2)) \oplus 1 = j \bmod 2$. Thus $F(\mathcal{C}_m) \subseteq \mathcal{C}_{m-1}$. For $m = 1$, the alternating left half is fixed, $x'_0 = 1 \oplus 1 = 0$, and $R'_1 = 0 \oplus (1 \vee R_2) = 1$ independently of the right tail; hence $F(\mathcal{C}_1) \subseteq \mathcal{C}_1$. Every $\mathcal{C}_m$ therefore realizes the zero trace.

A nonzero right half selects a realization by its first one, and Lemma 1 makes it unique; a zero right half forces a zero left half by the same lemma against the zero row.

The other constant trace has a simpler fiber: one row per right half, and the left half no longer depends on which right half it is.

Theorem 3 (All-one fiber). *For a prescribed right half with $R_0 = 1$, the left half compatible with an identically one trace is unique and is the same for every such right half.*

$$L_j = 1 \iff j \text{ is positive and even} \tag{8}$$

Proof. Write $L_0 := x_0 = R_0 = 1$, so that the third identity of (6), $L'_j = L_{j+1} \oplus (L_j \vee L_{j-1})$, applies at every $j \geq 1$. Under (8) it gives $0 \oplus (1 \vee 0) = 1$ for even j and $1 \oplus (0 \vee 1) = 0$ for odd j , so the left half is fixed by F whatever the right half is. At the origin, $F(x)_0 = L_1 \oplus (L_0 \vee R_1) = 0 \oplus 1 = 1$ independently of R_1 , so the trace is identically one and $L_0 = 1$ is restored at every step. Lemma 1 makes the left half unique.

3 Finite configurations

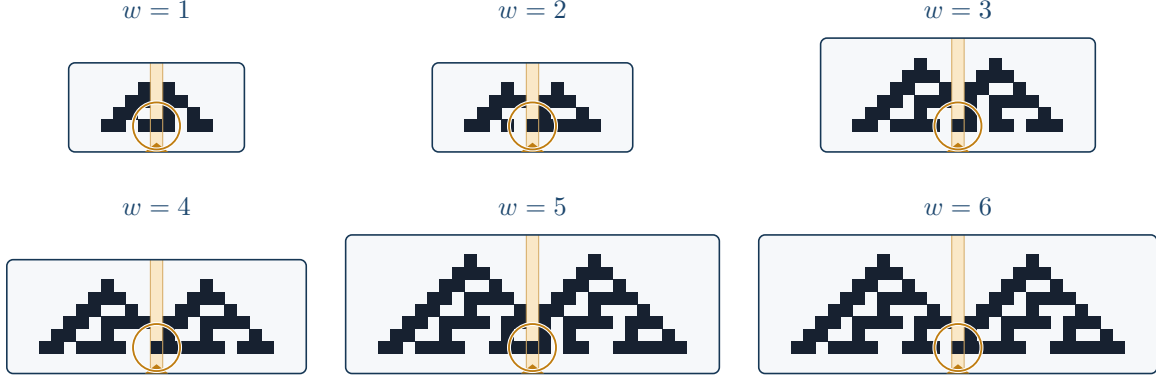
Every nonzero row in Theorem 2 carries ones arbitrarily far to the left, so none of them is finitely supported. That is the whole of the next statement, once one checks that a finite orbit cannot become the zero row.

Corollary 4. $\text{Tr}^{-1}(0^{\mathbb{N}}) \cap \mathcal{F} = \{0\}$. *Moreover, no column of the orbit of a nonzero $x \in \mathcal{F}$ is eventually zero.*

Proof. Every nonzero row in Theorem 2 has ones at all sufficiently large odd left depths. If a column of a finite orbit were zero from time T onward, translation would make $F^T(x)$ a finite zero-trace row. It is nonzero: if $[a, b]$ is the support interval of a nonzero finite row, its next row has ones at $a - 1$ and $b + 1$ because Rule 30 maps 001 and 100 to one.

Corollary 4 is not a consequence of left permutivity alone: Rule 90 is also left permutive, yet the finite row with ones at ± 1 keeps a zero center forever (Figure 1(b,c)). The Rule-30-specific step is the OR latch $R'_1 = 1 \vee R_2 = 1$ in the proof of Theorem 2, which has no analogue in an additive rule.

The same latch upgrades the corollary from one constant to both, with no further machinery. Theorem 3 proves the second half again and independently, since (8) has ones at every positive even depth; the argument below is shorter and is the one to quote.



circled cell = first nonzero center, at time $2\lceil w/2 \rceil + 1$

Figure 2: One extremizer of Theorem 6 per radius, truncated from the zero-trace completion at left depth w . The horizon is flat across each pair $w = 2k - 1, 2k$, which is the parity effect in the statement.

Corollary 5. *No column of the orbit of a nonzero $x \in \mathcal{F}$ is eventually constant.*

Proof. The eventually-zero case is Corollary 4. Suppose instead some column is eventually one; translating, take it to be column 0, so $c_t = 1$ for all $t \geq T$. Applying (1) at the origin gives $l_t = c_{t+1} \oplus (c_t \vee r_t)$, and $c_t = 1$ collapses the disjunction, so $l_t = 1 \oplus c_{t+1}$. For $t \geq T$ we also have $c_{t+1} = 1$, hence $l_t = 0$. Column -1 is then zero from time T onward, which Corollary 4 forbids.

4 A sharp finite horizon

Corollary 4 says a finite row's center cannot stay zero forever. How long it can stay zero, as a function of the support radius, is exactly one step past the light cone of the first odd depth beyond that radius.

For $w \geq 1$, let $\mathcal{S}_w = \{x \neq 0 : \text{supp}(x) \subseteq [-w, w], x_0 = 0\}$. For $x \in \mathcal{S}_w$, the zero-center horizon is defined as follows.

$$H(x) = \max\{h : F^t(x)_0 = 0 \text{ for } 0 \leq t \leq h\} \quad (9)$$

Thus $H(x)$ is the last matching time index; the corresponding prefix contains $H(x) + 1$ symbols, including the symbol at time zero.

Theorem 6 (Sharp finite horizon). *Let $w \geq 1$. Then both of the following hold.*

$$\max_{x \in \mathcal{S}_w} H(x) = 2 \left\lceil \frac{w}{2} \right\rceil, \quad \#\left\{x \in \mathcal{S}_w : H(x) = 2 \left\lceil \frac{w}{2} \right\rceil\right\} = 2^w - 1 \quad (10)$$

Proof. Let q be the least odd integer greater than w , so $q - 1 = 2\lceil w/2 \rceil$. If the positive right half is nonzero, a trace zero through time q , compared with its zero-trace completion, would force by finite-prefix uniqueness and (4) $x_{-q} = \bigvee_{j=1}^q R_j = 1$, contradicting $q > w$. If that half is zero, comparison with the zero row makes every nonzero left cell visible by time w . Thus $H(x) \leq q - 1$.

Conversely, choose any of the $2^w - 1$ nonzero words $(R_1, \dots, R_w)$, install (4)–(5) through left depth w , and zero all other cells. This row agrees with its infinite zero-trace completion through the light cone of time $q - 1$; when w is odd, the extra depth $w + 1$ is even and forced to zero. The missing odd cell at depth q creates a discrepancy at time q , and finite-prefix uniqueness excludes further extremizers.

The one-trace horizon is the same computation against (8), and the parities exchange roles: ones sit at even depths, so the binding depth is the least *even* one past the radius, and $R_0 = 1$ already makes the row nonzero, which is why no extremizer is lost here.

Theorem 7 (Sharp finite horizon, one). *Let $w \geq 1$, let $\mathcal{T}_w = \{x : \text{supp}(x) \subseteq [-w, w], x_0 = 1\}$, and for $x \in \mathcal{T}_w$ let $H_1(x) = \max\{h : F^t(x)_0 = 1 \text{ for } 0 \leq t \leq h\}$. Let q be the least even integer greater than w . Then both of the following hold.*

$$\max_{x \in \mathcal{T}_w} H_1(x) = q - 1, \quad \#\{x \in \mathcal{T}_w : H_1(x) = q - 1\} = 2^w \quad (11)$$

Proof. A trace one through time q , compared with the all-one completion of Theorem 3, would force by finite-prefix uniqueness and (8) $x_{-q} = 1$, contradicting $q > w$; thus $H_1(x) \leq q - 1$. Conversely, choose any of the 2^w words $(R_1, \dots, R_w)$, install (8) through left depth w , and zero all other cells. This row agrees with its all-one completion through the light cone of time $q - 1$; when w is even, the extra depth $w + 1$ is odd and forced to zero. The missing even cell at depth q creates a discrepancy at time q , and finite-prefix uniqueness excludes further extremizers.

Combining the two horizons, the sharp law over both constants is flat in the parity that splits each one separately.

Corollary 8. *Let $w \geq 1$ and let $x \neq 0$ have $\text{supp}(x) \subseteq [-w, w]$. The longest constant prefix of the central trace of x has length at most $w + 2$, and exactly 2^w configurations attain length $w + 2$ when w is even, while exactly $2^w - 1$ attain it when w is odd.*

Proof. For $x_0 = 0$ Theorem 6 gives $2\lceil w/2 \rceil$, which is w for even w and $w + 1$ for odd w ; for $x_0 = 1$ Theorem 7 gives $q - 1$, which is $w + 1$ for even w and w for odd w . Hence the maximum horizon index is $w + 1$, so the maximum prefix length is $w + 2$. It is attained in the one case when w is even and in the zero case when w is odd, with the corresponding counts 2^w and $2^w - 1$.

5 Scope and further work

Theorems 2 and 3 classify the two constant fibers of the trace map, and Corollaries 4 and 5 say that neither meets the finite configurations away from the origin. The next unresolved case is eventual period two; no claim about that case or any higher period is made here.

6 Reproducibility and formalization

The accompanying source archive was regenerated against repository snapshot

99b2d82af113e385f43452eb2b8f1b22180e6de0.

From the repository root, the following commands reproduce the computational checks and figures.

```
cd docs/rule30/paper
uv run python ../../../../experiments/rule30/constant_trace_audit.py
uv run python ../../../../experiments/rule30/figures.py --check --out figures
uv run --with z3-solver python \
  ../../../../experiments/rule30/zero_tail_smt_certificate.py --json
latexmk -pdf -interaction=nonstopmode -halt-on-error zero-tail-note.tex
```

The exhaustive audit must end with **FAILURES: none**. For $w = 1, \dots, 7$, its zero-horizon/count pairs are 2/1, 2/3, 4/7, 4/15, 6/31, 6/63, 8/127; its one-horizon/count pairs are 1/2, 3/4, 3/8, 5/16, 5/32, 7/64, 7/128. These are horizon indices, so the associated prefix lengths are one larger. The figure command must print **figures: all claimed properties hold**; it generates the three panels of Figure 1 and the six panels of Figure 2. The SMT report must return status **PASS** with all 18 local Boolean obligations **unsat**. The paper has no numerical tables. A successful L^AT_EX build produces a six-page PDF.

The Lean 4 file `experiments/rule30/Rule30ZeroTail.lean` is explicitly partial. It formalizes the local Rule 30 equation and left permutivity, arbitrary-depth triangular sensitivity, the prefix-OR formula conditional on a first right-hand one, the Boolean identities used for $F(\mathcal{C}_m) \subseteq \mathcal{C}_{m-1}$ and $F(\mathcal{C}_1) \subseteq \mathcal{C}_1$, and the infinite-left-tail contradiction conditional on the classification. It does not formalize the derivation of that classification from a zero trace, the all-one fiber, either sharp horizon theorem, or the eventual-constant corollaries. With Lean 4.33.1 installed, it is checked by

```
lean ../../../../experiments/rule30/Rule30ZeroTail.lean
```

which should exit successfully without output. Lean was not installed in the environment used for the present editorial rebuild, so that command was not rerun here. Every cell in both figures was regenerated directly from (1); the exhaustive and SMT checks supplement, but do not replace, the unbounded proofs above.

References

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